

AI Ecommerce Reporting Agent 14 Advanced

n8n + OpenAI API + Shopify + Dual-Stage AI Pipeline

Automated ecommerce KPI reporting with a dual-stage AI pipeline:
Stage 1 generates management reports, Stage 2 classifies business health (HEALTHY / WARNING / CRITICAL) and auto-routes actions.

n8n

OpenAI API

Shopify API

KPI Engine

Health Monitor

Email

Dashboard

KPI

OVERVIEW

Project Summary

This advanced AI agent automates ecommerce performance reporting end-to-end. A scheduled **n8n workflow** pulls order data from **Shopify**, calculates six core KPIs, sends them to **OpenAI GPT-4o (Stage 1)** for a natural-language management report, then passes the results to a **second AI stage** that classifies overall business health as **HEALTHY**, **WARNING**, or **CRITICAL**. A Switch node routes each classification to the appropriate automated business action — from scaling campaigns to triggering an immediate management review.

TRACKED KPIs

Revenue Total order value in the reporting period.	Orders Total number of completed orders.	AOV Average Order Value = Revenue / Orders.
Conv. Rate % of site visitors who completed a purchase.	ROAS Return on Ad Spend = Revenue / Ad Spend.	Refund Rate % of orders that resulted in a refund.

ENHANCED ARCHITECTURE

Dual-Stage AI Pipeline

This workflow uses **two sequential AI calls** — a design pattern central to advanced AI agent architecture. Stage 1 is generative (report writing); Stage 2 is evaluative (business health classification). Both share the same KPI context.

1 Shopify Data Order source	2 n8n Scheduler Cron trigger	3 KPI Processing Calculate metrics	4 AI Stage 1 Report generation	5 AI Stage 2 Health classifier	6 Switch Node Route by health	7 Auto Actions Scale / Review / Alert
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Nodes 4 and 5 (highlighted) form the dual-stage AI core. Stage 1 writes the report; Stage 2 reads the report + raw KPIs and issues the health classification.

IMPLEMENTATION

Step-by-Step Execution

Step 1	Schedule Workflow An n8n Schedule Trigger fires the workflow at a defined interval (daily at 08:00, weekly on Monday, or end-of-month). No manual intervention required. The cron expression is fully configurable without code changes.
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Step 2	Pull Order Data from Shopify The Shopify API node fetches all orders for the reporting period — order IDs, total price, refund status, timestamps, and customer data. Sample data or a CSV node can be used as a mock source during development and demos.
Step 3	Calculate KPIs A Function/Set node computes all six KPIs from the raw order data: Total Revenue, Order Count, AOV (Revenue / Orders), Conversion Rate (orders / sessions from analytics), ROAS (Revenue / Ad Spend), and Refund Rate (refunded orders / total orders x 100).
Step 4	AI Stage 1 — Generate Management Report KPI values are injected into a GPT-4o prompt instructing the model to write a concise executive summary: performance highlights, week-over-week trends, and 3 recommended actions. Output is a clean narrative paragraph in plain English.
Step 5	AI Stage 2 — Business Health Classification A second OpenAI call receives the KPI values + the Stage 1 report. The system prompt instructs GPT to return ONLY one label: HEALTHY, WARNING, or CRITICAL, based on defined KPI thresholds (e.g. ROAS < 1.5 = WARNING, Refund Rate > 8% = CRITICAL).
Step 6	Switch Node — Route by Health Status The Switch node reads the classification label and activates one of three branches: HEALTHY routes to scaling actions, WARNING triggers a metrics review alert, CRITICAL sends an immediate management escalation email.
Step 7	Email Report + Dashboard Update All branches send a formatted email with the management report, KPI table, and health status badge. Optionally, a Google Sheets or dashboard node is updated with the KPI snapshot for historical trend tracking.

AI HEALTH CLASSIFICATION

HEALTHY / WARNING / CRITICAL Routing

Stage 2 of the AI pipeline evaluates overall business performance against KPI thresholds and returns one of three health labels. The Switch node then dispatches the correct automated business action.

HEALTHY

KPI Thresholds: ROAS >= 2.0, Conversion Rate >= 3%, Refund Rate <= 3%, Revenue on target or above.

Recommended Action: Continue scaling campaigns, increase ad budget allocation, maintain current strategy.

Action: Send weekly summary email. Flag for growth team review. No escalation needed.

WARNING

KPI Thresholds: ROAS 1.5-2.0, Conversion Rate 1.5-3%, Refund Rate 3-8%, Revenue slightly below target.

Recommended Action: Review marketing spend efficiency, analyse checkout funnel drop-offs, audit top SKUs.

Action: Send alert to marketing manager. Schedule weekly performance review call.

CRITICAL

KPI Thresholds: ROAS < 1.5, Conversion Rate < 1.5%, Refund Rate > 8%, or revenue > 20% below target.

Recommended Action: Immediate management review. Pause underperforming ad campaigns. Investigate refund spike.

Action: Escalation email to C-level. Freeze ad spend increases. Trigger audit workflow.

DEMO SCENARIO

End-to-End Live Example

The n8n scheduler fires on Monday 08:00. Here is the complete workflow trace from raw Shopify data to the automated business action:

KPI Snapshot	Revenue: \$48,200 Orders: 312 AOV: \$154.49 Conv. Rate: 1.8% ROAS: 1.6 Refund Rate: 5.2%
AI Stage 1 — Report	Revenue is 12% below weekly target. ROAS of 1.6 indicates marginal ad efficiency. Refund rate trending upward – requires product quality review. Recommend: audit top-3 SKUs, pause bottom-performing ad sets.
AI Stage 2 — Health	{"health": "WARNING", "reason": "ROAS below 2.0 threshold, conversion rate underperforming at 1.8%, refund rate approaching alert level of 5%."}
Switch Routing	WARNING branch activated -> Email to: marketing-team@company.com Subject: [WARNING] Weekly Ecommerce Health Report – Action Required Body: KPI table + AI report + 3 recommended actions attached.

TOOLS USED

Technology Stack

n8n	Workflow engine. Schedule Trigger, HTTP, Function, Switch, Email, and Google Sheets nodes used throughout.
OpenAI API (x2)	Two sequential GPT-4o calls: Stage 1 for report generation, Stage 2 for health classification with JSON output.
Shopify API	Pulls order data for the reporting window. A CSV or mock data node can substitute during development.
KPI Engine	n8n Function node that computes Revenue, Orders, AOV, Conversion Rate, ROAS, and Refund Rate from raw data.
Switch Node	Routes workflow into HEALTHY, WARNING, or CRITICAL branches based on the Stage 2 classification label.
Email Node	Sends formatted reports to the relevant team. Content varies by health tier — standard, alert, or escalation.
Google Sheets	Optional: stores KPI snapshots over time for trend analysis and historical performance dashboards.

INTERVIEW TALKING POINTS

Key Topics to Present

- **Ecommerce Analytics & KPI Design**
Explain the six KPIs selected and why each matters: Revenue and Orders for volume, AOV for basket efficiency, Conversion Rate for funnel health, ROAS for ad ROI, and Refund Rate as a product quality / satisfaction proxy.
- **Dual-Stage AI Agent Architecture**
This is the most technically differentiated aspect. Stage 1 is generative — it produces a narrative report. Stage 2 is evaluative — it reads the report and raw KPIs to issue a structured health verdict. Two LLM calls, two distinct responsibilities.
- **AI-Powered Business Health Monitoring**
The system moves beyond passive reporting into active monitoring. The health classification layer enables the workflow to make conditional business decisions autonomously — a core characteristic of AI agent design patterns.
- **Automated Decision-Making Workflows**
Emphasise the Switch node routing: different health states trigger different automated actions, from growth scaling to C-level escalation. The workflow is not just a report generator — it is an operational decision loop.

- **Reporting Efficiency & Business Impact**

Manual weekly reporting typically takes 2-4 hours per analyst. This workflow generates the same output in under 60 seconds, every week, with zero human effort — freeing the analytics team for higher-value work.

- **KPI-Based Risk Assessment**

Highlight that the thresholds (ROAS < 1.5, Refund Rate > 8%) are configurable business rules injected into the AI prompt — not hardcoded. Non-technical stakeholders can adjust sensitivity without touching the workflow.

- **Scalability & Multi-Channel Extension**

The same architecture can ingest data from Amazon, WooCommerce, or any platform with an API. Additional KPI dimensions (LTV, CAC, Churn Rate) can be added to the KPI node and prompt with no structural changes.

EXTENSIBILITY & MIGRATION NOTE

Both AI stages use **OpenAI GPT-4o** and can be individually migrated to **Claude API (Anthropic)** or any OpenAI-compatible endpoint. The Shopify source can be replaced with **WooCommerce, Amazon SP-API, or a Google Sheets CSV export** by swapping only the data input node. Additional health tiers (e.g. RECOVERING, SCALING) can be added by extending the Switch node and updating the Stage 2 classification prompt — no pipeline restructuring required.